Development of a Data-Assimilative Digital Twin for Wave Energy Converters Using Field, Laboratory and Numerical Data

Abstract

This project aims to develop a data-assimilative digital twin framework for Wave Energy Converters (WEC), by integrating physics-based hydrodynamic models with PacWave field measurements, physical tank-test data from multiple WEC devices (subWEC, LUPA, FOSWEC) tested at the Hinsdale Wave Research Laboratory and PTO control-system data from the WESRF laboratory. Leveraging the existing computational workflows, calibrate and validate numerical models and the requested computing resources; the project will assimilate measurements to update model states and parameters and quantify predictive uncertainty. Key outputs will include WEC motion, PTO response estimates, power-production predictions and model-performance diagnostics. The framework will establish a computational foundation for improved WEC performance assessment, predictive operation and future control optimization under realistic ocean conditions.

1 Project Overview and Research Opportunity

This project proposes to develop a data-assimilative digital twin for wave energy converters (WECs) that connect ocean wave measurements, laboratory experiments and numerical simulations to predict device response and power conversion across scales. The framework shown in Fig. 1 will combine full-scale wave data from PacWave, scaled WEC hydrodynamic experiments at Oregon State University's O.H. Hinsdale Wave Research Laboratory and power take-off (PTO) bench experiments at WESRF to create a comprehensive model of WEC behavior. Measured wave conditions from PacWave will be scaled to laboratory conditions and used as inputs to physics-based WEC models to predict device motions and hydrodynamic forces. Measurements from Hinsdale wave-tank experiments and WESRF PTO tests will then be combined with the numerical predictions through data assimilation to improve estimates of device response, calibrate selected model parameters and assess prediction uncertainty. Building on existing numerical tools and WESRF's force/position driven and coupled testing capabilities, the project will establish and validate a generalized, reusable workflow linking tank testing, hydrodynamic simulation with physical PTO testing. The resulting framework will support assessment of loads on devices, dynamics and power output under representative ocean conditions. This workflow will provide a foundation for future integration of operational measurements from deployed devices.

2 Previous Research and Integration Gap

Digital twins are virtual representations of physical systems that can be used to simulate and optimize their behavior. The development of digital twins for wave energy converters (WECs) has

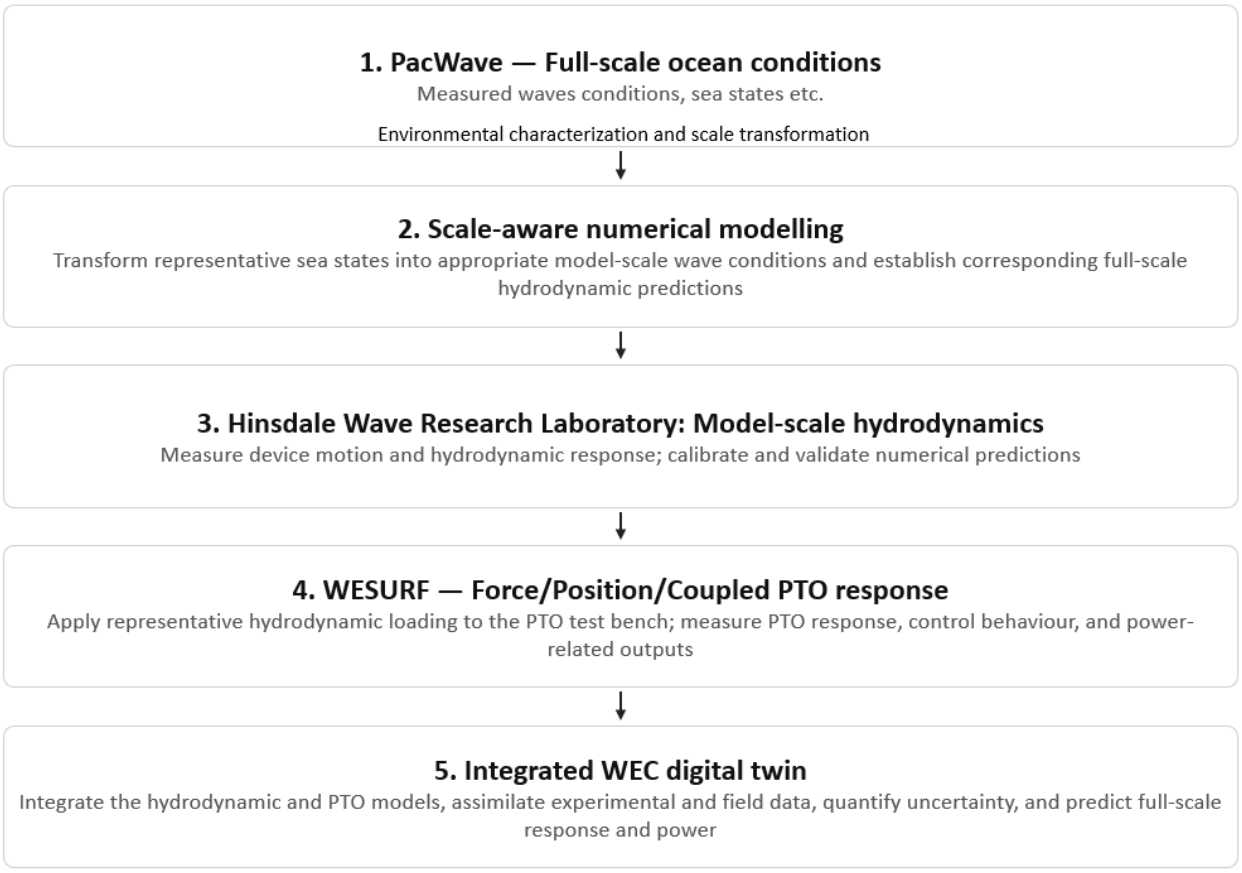


Figure 1: Framework for the proposed data-assimilative digital twin for wave energy converters.

gained significant attention in recent years due to the increasing need for accurate and reliable prediction of device performance under real-world conditions. A digital twin can help in understanding the complex interactions between the ocean environment, the WEC itself, and its control systems. This understanding is crucial for improving the design, operation and maintenance of WECs.

In a digital twin framework experimental data is used to calibrate and validate numerical models, which can then be used to predict the behavior of the physical system under different conditions. PacWave PacWave (2026) wave resource data was characterized by Dunkle et al. (2020) and Robertson et al. (2022) examined how spectral conditions at PacWave and other sites influence device-size-dependent energy production and variability. At field scale, Okushemiya et al. (2026) combined deployment measurements with PTO dynamometer data to improve modelling of a tethered-heave-plate WEC, incorporating generator inertia and measured power-conversion behaviour. Ali et al. (2025, 2026) in several tank-testing campaigns at OSU investigated the behavior of submerged-WEC's under various conditions. Nonlinear surge–pitch coupling and subharmonic excitation was investigated respectively in these studies, highlighting the importance of considering nonlinear effects in WEC modeling. McConnell et al. (2025, 2026) developed and experimentally evaluated real-time hybrid approaches for WEC mooring dynamics using Hinsdale Wave Research Laboratory and PTO bench experiments at WESRF. Additionally, the OSU–Sandia–Evergreen FOSTWIN Sandia National Laboratories et al. (2026) platform offers both a physics-based WEC-Sim model and a model identified from Hinsdale experimental data for controller development. Collectively, these studies provide the data and methodologies to motivate the proposed integration of

PacWAVE field wave conditions, tank-validated hydrodynamics and physical PTO testing within a reusable data-assimilation workflow.

Outside the Pacific Marine Energy Center (PMEC), similar efforts are being made to develop digital twins for wave energy converters. At ocean scale, Dizon et al. (2024) evaluated WEC-Sim and Gazebo models of the deployed MBARI two-body WEC against field measurements across 56 observations, comparing piston velocity, motor speed, and electrical power; the Gazebo model was intended to serve as a digital twin. Previous research provides complementary foundations for WEC digital twins through experimental model identification, field validation, and coupled PTO testing. Giorgi et al. (2019) used six experimental datasets to identify and validate efficient linear and nonlinear hydrodynamic models, demonstrating how laboratory measurements can inform predictive models. Lawal et al. (2026) discuss AI-driven digital twins for wave energy systems, while Katsidoniotaki et al. (2022) developed a CFD-derived surrogate for rapid prediction of extreme mooring loads. Related computational approaches include nonlinear response prediction using multiple-input operator networks by Zhang et al. (2025), CFD-coupled reinforcement learning for adaptive PTO control by Liang et al. (2024), and data-driven control methods reviewed by Pasta et al. (2023). For physical PTO testing, Bracco et al. (2015) coupled an ISWEC gyroscope and generator test rig with a numerical hydrodynamic model through hardware-in-the-loop simulation, while Bacelli et al. (2019) demonstrated control testing through complementary dry-bench and wave-tank experiments. Together, these studies motivate a framework connecting measured ocean-wave conditions, tank-validated hydrodynamics, and physical PTO response.

The proposed project will build on these approaches by developing a reusable workflow that explicitly assimilates laboratory measurements to update model estimates and assess prediction uncertainty across the hydrodynamic and PTO testing stages.

3 Research Objectives and Expected Outcomes

The project has five linked objectives.

1. Identify the representative wave conditions derived from PacWave measurements.
2. Organize the existing tank-testing and PTO (Pneumatic) datasets into a documented structure that preserves device geometry, scale, instrumentation, wave conditions and PTO settings.
3. Identify the most appropriate numerical models (WEC-Sim, CFD) that reproduce selected laboratory experiments and validate their predictions against independent datasets.
4. Develop and evaluate a data-assimilation method that uses measured device response to update estimated states and selected model parameters.
5. Integrate the resulting hydrodynamic model with WESRF's physical PTO testing capabilities and evaluate the coupled response.

Expected outcomes include a curated dataset inventory, validated baseline and updated models, a documented coupled simulation and testing workflow and an assessment of prediction accuracy and uncertainty. Success will be demonstrated through reproducible comparisons of device motion, PTO displacement or velocity, measured connection loads and power metrics.

4 Research Approach: Connecting Field, Tank, and PTO Data

Continuous PacWave wave measurements will define representative environmental conditions for numerical simulations and laboratory comparisons with Froude scaling applied to the relevant

scales. Existing tank datasets from Ali et al. (2026); Beringer et al. (2025, 2026) and collaborators will support hydrodynamic model calibration and independent validation. Each dataset will retain its original geometry, scale, and test configuration, and simulations will use measured laboratory waves wherever available. Comparisons will examine wave parameters, relative body motion and measured PTO quantities. Ali's studies will inform assessment of nonlinear coupled motions, while Beringer's uncertainty methods will help distinguish model discrepancies from experimental variability.

PTO modelling and testing will build on Inyong Kim's previous WESRF work and the laboratory's existing force/position and coupled testing capabilities. Available bench records will characterize PTO behavior. Numerical hydrodynamics and the physical PTO exchange force or motion information will be exchanged. McConnell and Bosma's hybrid simulation experience will inform communication, actuator tracking, and delay assessment. The resulting interfaces and procedures will be documented as a reusable workflow.

Data assimilation will first be evaluated using recorded laboratory measurements before integration with live bench testing. A state estimator will combine model predictions with motion and PTO measurements to update estimated positions and velocities. Measurement and model uncertainty will inform the updates. Performance will be assessed against uncalibrated and calibrated models without assimilation, using independent cases to evaluate

NVIDIA Omniverse will be evaluated as a common visualization environment connecting WEC geometry, experimental measurements, and numerical predictions. An OpenUSD representation of the device and selected test configurations will support synchronized replay of existing tank and PTO bench datasets alongside simulated responses.

5 Computational Readiness and Resource Requirements

Computational readiness is dependent on several years of numerical workflows, accessible experimental datasets and established laboratory hardware. Available modelling tools include frequency-domain, time-domain WEC-Sim, ANSYS-Aqwa and ProteusDS workflows and OpenFOAM CFD capabilities, supported by MATLAB and Python processing tools. Existing tank datasets from testing of LUPA, SubWEC and FOSSWEC and prior WESRF testing reduce dependence on a new experimental campaign to initiate the research. The first project stage will document dataset availability, calibration records and model validations. Advanced computing will support parallel hydrodynamic parameter studies, selected CFD investigations of nonlinear response and ensemble calculations used to examine uncertainty. Efficient linear models will support repeated estimation and coupled testing, while higher-fidelity calculations will be targeted at cases where simpler models show unresolved discrepancies. Computing requirements will be estimated through benchmark runs, including processor time, memory and storage, before larger workloads are scheduled.

6 Work Plan, Milestones, and Validation

The project will be completed over twelve months. Proposed activities and milestones are summarized in Table 1. The first four months will focus on dataset inventory, model configuration, and baseline validation. Months five through eight will develop and evaluate the data-assimilation framework, including parameter identifiability and uncertainty assessment. The final four months will integrate the hydrodynamic model with WESRF PTO testing, validate the coupled workflow, and assess full-scale predictions.

Table 1: Monthly project activities, milestones, and three research phases.

Month	Phase	Main activities	Milestone / deliverable
1	Phase I	Inventory and quality-check PacWave, Hinsdale, and WESURF datasets; select test cases.	Dataset inventory and calibration–validation split.
2		Configure hydrodynamic models using measured tank-wave inputs.	Reproducible baseline simulations.
3		Compare predicted and measured motions, spectra, and available PTO quantities.	Baseline accuracy and discrepancy assessment.
4		Calibrate selected parameters and evaluate withheld tank cases.	Baseline model validation report.
5	Phase II	Develop a state estimator using recorded laboratory measurements.	Initial data-assimilation implementation.
6		Evaluate estimation and prediction accuracy; assess parameter identifiability.	Comparison with models without assimilation.
7		Perform targeted CFD and ensemble studies to assess uncertainty.	Model and prediction uncertainty assessment.
8		Evaluate assimilation on independent cases and characterize PTO behaviour using bench records.	Evaluated assimilation framework and coupling requirements.
9	Phase III	Integrate the hydrodynamic model with WESURF interfaces; conduct recorded-input tests.	Verified communication, timing, and actuator tracking.
10		Conduct coupled simulation–PTO tests and independent validation.	Validated coupled workflow and repeatability assessment.
11		Simulate representative PacWave sea states; assess uncertainty and scaling limitations.	Laboratory-informed full-scale predictions.
12		Finalize software, documentation, datasets, and manuscripts.	Reusable workflow, final report, and manuscripts.

7 Impact Readiness and Pathway to External Funding

The project will increase the scientific value of existing investments in PacWave monitoring, Hinsdale tank experiments, and WESURF PTO testing by connecting their outputs within a reproducible modelling framework. Its immediate contribution will be improved understanding of how hydrodynamic response, PTO behaviour, and measurement uncertainty influence WEC performance predictions. A reusable workflow could reduce repeated model-setup effort and help researchers identify where additional experiments or higher-fidelity computations are most valuable. The project will also provide practical training in marine hydrodynamics, scientific computing, experimental data processing, state estimation, and hardware–software integration. Documented examples and datasets suitable for release will support student research and future capstone activities, subject to applicable data-access arrangements.

The validated workflow and initial results will provide evidence for subsequent proposals to DOE water-power programs, NSF, ONR, and other relevant sponsors. Future research could extend the

framework to deployed-device measurements, additional PTO technologies, controller evaluation, and load monitoring. This award will establish the computational and experimental foundation needed to pursue those applications credibly. Advanced computing is central to generating the parameter and uncertainty ensembles and selected high-fidelity simulations that would be impractical to obtain through physical testing alone. Coupling those calculations with existing measurements will produce a stronger basis for future funding proposals, publications, and collaborations focused on reliable wave-powered ocean systems.

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